# Chapter 2. Topology of Euclidean Space

Open and closed sets, Bolzano–Weierstrass, Cauchy sequences, series, compactness, connectedness

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## 2.1 Euclidean space

**Motivation.** To study functions of several variables we need distance in $\mathbb{R}^n$. The Cauchy–Schwarz inequality is what makes the Euclidean distance satisfy the triangle inequality.

$\mathbb{R}^n$ is the set of $n$-tuples $x = (x_1, \dots, x_n)$ with coordinatewise addition and scalar multiplication, the **inner product** $\langle x, y\rangle = \sum x_i y_i$, the **norm** $\|x\| = \langle x, x\rangle^{1/2}$ and the **distance** $\|x - y\|$.

**Theorem 2.1 (2.1.1, Cauchy–Schwarz inequality)**

$|\langle x, y\rangle| \le \|x\|\|y\|$ for all $x, y \in \mathbb{R}^n$, with equality iff $x, y$ are linearly dependent. Consequently $\|x + y\| \le \|x\| + \|y\|$ and $\|x - z\| \le \|x - y\| + \|y - z\|$.

**Proof.** If $y = 0$ both sides vanish. Otherwise, for real $t$, $0 \le \|x - ty\|^2 = \|x\|^2 - 2t\langle x, y\rangle + t^2\|y\|^2$; take $t = \langle x, y\rangle / \|y\|^2$ to get $0 \le \|x\|^2 - \langle x, y\rangle^2/\|y\|^2$. Equality forces $x = ty$. Then $\|x + y\|^2 = \|x\|^2 + 2\langle x, y\rangle + \|y\|^2 \le (\|x\| + \|y\|)^2$. ∎

Also $\max_i |x_i| \le \|x\| \le \sqrt{n}\max_i |x_i|$, so a sequence in $\mathbb{R}^n$ converges iff each coordinate sequence converges.

## 2.2 Open and closed sets

**Motivation.** Many theorems need a domain without boundary points (to differentiate) or one containing all its limits (to guarantee maxima). Open and closed sets express these two requirements.

The **$\epsilon$-neighbourhood** of $x$ is $N(x, \epsilon) = \{y : \|y - x\| < \epsilon\}$. A point $x$ is an **interior point** of $A$ if some $N(x, \epsilon) \subset A$; $A$ is **open** if all its points are interior. $x$ is a **limit point** of $A$ if every $N(x, \epsilon)$ contains a point of $A$ other than $x$; $A'$ denotes the set of limit points, and $\overline{A} = A \cup A'$ is the **closure**. $A$ is **closed** if $\mathbb{R}^n - A$ is open. Points of $A - A'$ are **isolated points**. Every $N(x, \epsilon)$ is open, by the triangle inequality.

**Proposition 2.2 (2.2.1)**

(a) $\emptyset$ and $\mathbb{R}^n$ are open;
(b) unions of open sets are open;
(c) finite intersections of open sets are open. Dually, $\emptyset, \mathbb{R}^n$ are closed, intersections of closed sets are closed, and finite unions of closed sets are closed.

**Proof.** (a), (b) are immediate.

(c) If $x \in U_1 \cap \cdots \cap U_k$ with $N(x, \epsilon_i) \subset U_i$, then $N(x, \min \epsilon_i)$ lies in the intersection. The dual statements follow by De Morgan's laws. ∎

An infinite intersection of open sets need not be open: $\bigcap_n (-1/n, 1/n) = \{0\}$.

**Proposition 2.3 (2.2.2)**

For $A \subset \mathbb{R}^n$ and $x \in \mathbb{R}^n$:
(a) $x \in A'$ iff some sequence in $A - \{x\}$ converges to $x$;
(b) $x \in \overline{A}$ iff some sequence in $A$ converges to $x$.

**Proof.** (a) If $x \in A'$, pick $x_k \in A \cap N(x, 1/k)$, $x_k \ne x$. Conversely, if $x_k \to x$ with $x_k \in A - \{x\}$, every $N(x, \epsilon)$ contains some $x_k$.

(b) If $x \in A$ use the constant sequence; otherwise $x \in A'$ and use (a). Conversely, if $x_k \to x$, $x_k \in A$, either $x = x_k$ for some $k$ or the $x_k$ lie in $A - \{x\}$. ■

**Theorem 2.4 (2.2.3)**

For $F \subset \mathbb{R}^n$ the following are equivalent:
(a) $F$ is closed;
(b) $F' \subset F$;
(c) $F = \overline{F}$.

**Proof.** (a)⇒(b): if $x \notin F$, some $N(x, \epsilon)$ misses $F$, so $x \notin F'$. (b)⇒(c): $\overline{F} = F \cup F' = F$. (c)⇒(a): if $x \notin F$, then $x \notin F'$, so some $N(x, \epsilon)$ contains no point of $F$ (other than $x$, which is not in $F$); thus $\mathbb{R}^n - F$ is open. ■

**Corollary 2.5 (2.2.4)**

$A'$ is closed for every $A \subset \mathbb{R}^n$; hence $\overline{A}$ is closed, and it is the smallest closed set containing $A$.

**Proof.** Let $x \in (A')'$ and $\epsilon > 0$. Choose $y \in A'$ with $0 < \|y - x\| < \epsilon$, and $\delta = \min(\|y - x\|, \epsilon - \|y - x\|) > 0$. Then $N(y, \delta)$ contains a point $z \in A$ with $z \neq y$; $z \in N(x, \epsilon)$, and $z \neq x$ because $\|z - y\| < \|x - y\|$. So $x \in A'$. The same argument shows $(\overline{A})' \subset A'$: if $N(x, \epsilon)$ contains $y \in A \cup A'$ with $y \neq x$, either $y \in A$, or $y \in A'$ and a small neighbourhood of $y$ inside $N(x, \epsilon)$ contains a point of $A$ other than $x$. So $\overline{A}$ is closed by Theorem 2.4. If $F \supset A$ is closed, $F \supset F' \supset A'$, so $F \supset \overline{A}$. ■

For $X \subset \mathbb{R}^n$, a set $U \subset X$ is **open in** $X$ (relatively open) if for each $x \in U$ there is $\epsilon > 0$ with $N(x, \epsilon) \cap X \subset U$.

**Proposition 2.6 (2.2.5)**

For $U \subset X \subset \mathbb{R}^n$ the following are equivalent:
(a) $U$ is open in $X$;
(b) $U = V \cap X$ for some open $V \subset \mathbb{R}^n$.

**Proof.** (a)⇒(b): for $x \in U$ choose $\epsilon_x$ with $N(x, \epsilon_x) \cap X \subset U$, and let $V = \bigcup_x N(x, \epsilon_x)$. (b)⇒(a): if $x \in V \cap X$ and $N(x, \epsilon) \subset V$, then $N(x, \epsilon) \cap X \subset U$. ■

Similarly, $F \subset X$ is closed in $X$ iff $F = E \cap X$ with $E$ closed in $\mathbb{R}^n$.

## 2.3 The Bolzano–Weierstrass theorem and Cauchy sequences

**Motivation.** Every bounded sequence has a convergent subsequence. This is the main way to produce limits out of nothing, and it shows that a sequence converges exactly when its terms eventually stay close to each other (the Cauchy criterion), which can be checked without knowing the limit.

A **box** is $B = I_1 \times \cdots \times I_n$ with bounded closed intervals $I_k$.

**Lemma 2.7 (2.3.1, nested boxes)**

If $B_1 \supset B_2 \supset \cdots$ are boxes, then $\bigcap_m B_m \neq \emptyset$.

**Proof.** Write $B_m = I_1^m \times \cdots \times I_n^m$. For each $k$, the intervals $I_k^m$ are nested, so contain a common point $x_k$ (Theorem 1.20); then $(x_1, \ldots, x_n) \in \bigcap B_m$. ■

**Theorem 2.8 (2.3.2, Bolzano–Weierstrass)**

Every bounded infinite set $A \subset \mathbb{R}^n$ has a limit point.

**Proof.** $A$ lies in a box $B_1$. Divide $B_1$ into $2^n$ boxes by bisecting each side; one of them, $B_2$, contains infinitely many points of $A$. Repeating, we obtain nested boxes $B_m$ whose sides have length $2^{1-m}$ times those of $B_1$, each containing infinitely many points of $A$. Let $x \in \bigcap B_m$ (Lemma 2.7). Any $N(x, \epsilon)$ contains $B_m$ for large $m$ (its diameter tends to 0), hence infinitely many points of $A$; so $x \in A'$. ■

**Proposition 2.9 (2.3.3)**

If $x$ is a limit point of $A$, then $A \cap N(x, \epsilon)$ is infinite for every $\epsilon > 0$.

**Proof.** If $A \cap N(x, \epsilon) - \{x\} = \{a_1, \ldots, a_k\}$, then $N(x, \delta)$ with $\delta = \min \|a_i - x\|$ contains no point of $A$ other than $x$, a contradiction. ■

**Theorem 2.10 (2.3.4)**

Every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.

**Proof.** If the range $\{x_k\}$ is finite, some value is taken infinitely often, giving a constant subsequence. Otherwise it is a bounded infinite set with a limit point $x$ (Theorem 2.8). Each $N(x, 1/j)$ contains infinitely many terms (Proposition 2.9), so we can choose $k_1 < k_2 < \cdots$ with $\|x_{k_j} - x\| < 1/j$. ■

For a bounded real sequence $(a_n)$ let $A$ be the set of limits of its convergent subsequences.

**Theorem 2.11 (2.3.5)**

$\limsup a_n = \max A$ and $\liminf a_n = \min A$.

**Proof.** Let $a = \limsup a_n$. By Proposition 1.17(b) we may choose $n_1 < n_2 < \cdots$ with $a_{n_k} > a - 1/k$, and by (a) of the same proposition $a_{n_k} < a + 1/k$ for large $k$; so $a_{n_k} \to a$ and $a \in A$. If $a_{n_k} \to b$, then $a_{n_k} < a + \epsilon$ for large $k$, so $b \le a$. The $\liminf$ case is symmetric. ■

A sequence is **Cauchy** if for every $\epsilon > 0$ there is $N$ with $\|x_m - x_n\| < \epsilon$ for all $m, n \ge N$.

**Proposition 2.12 (2.3.6)**

Every convergent sequence is Cauchy, and every Cauchy sequence is bounded.

**Proof.** If $x_n \to x$, choose $N$ with $\|x_n - x\| < \epsilon/2$ for $n \ge N$; then $\|x_m - x_n\| < \epsilon$. If $(x_n)$ is Cauchy, choose $N$ for $\epsilon = 1$; then $\|x_n\| \le \max\{\|x_1\|, \ldots, \|x_N\|\} + 1$. ■

**Lemma 2.13 (2.3.7)**

If a subsequence of a Cauchy sequence $(x_n)$ converges to $x$, then $(x_n)$ converges to $x$.

**Proof.** Given $\epsilon$, choose $N$ with $\|x_m - x_n\| < \epsilon/2$ for $m, n \ge N$, and $k$ with $n_k \ge N$ and $\|x_{n_k} - x\| < \epsilon/2$. For $n \ge N$, $\|x_n - x\| \le \|x_n - x_{n_k}\| + \|x_{n_k} - x\| < \epsilon$. ■

**Theorem 2.14 (2.3.8, Cauchy criterion)**

Every Cauchy sequence in $\mathbb{R}^n$ converges.

**Proof.** It is bounded (Proposition 2.12), so has a convergent subsequence (Theorem 2.10); apply Lemma 2.13. ■

**Corollary 2.15 (2.3.9, Cauchy criterion for series)**

$\sum a_n$ converges iff for every $\epsilon > 0$ there is $N$ with $|a_{m+1} + \cdots + a_n| < \epsilon$ for all $n > m \geq N$.

**Proof.** Apply Theorem 2.14 to the partial sums $s_n$, since $s_n - s_m = a_{m+1} + \cdots + a_n$. ■

In particular, if $\sum a_n$ converges then $a_n \to 0$, and if $\sum |a_n|$ converges (**absolute convergence**) then $\sum a_n$ converges, because $|a_{m+1} + \cdots + a_n| \leq |a_{m+1}| + \cdots + |a_n|$.

## 2.4 Series

**Motivation.** Infinite sums are limits of partial sums, so convergence of series is a special case of convergence of sequences. The tests below (comparison, ratio, root, alternating) are practical forms of the Cauchy criterion, and absolute convergence explains when terms may be rearranged.

**Proposition 2.16 (2.4.1, rearrangement of nonnegative series)**

Let $a_n \geq 0$ and $\tau : \mathbb{N} \to \mathbb{N}$ a bijection. If $\sum a_n = s$, then $\sum a_{\tau(n)} = s$; if $\sum a_n$ diverges, so does $\sum a_{\tau(n)}$.

**Proof.** Each partial sum $\sum_{n \leq N} a_{\tau(n)}$ is at most $\sum_{n \leq M} a_n \leq s$ with $M = \max \tau(\{1, \ldots, N\})$, so $t = \sum a_{\tau(n)} \leq s$ by Corollary 1.16. Applying this to $\tau^{-1}$ gives $s \leq t$. The divergence statement follows by symmetry. ■

More generally, an absolutely convergent series may be rearranged without changing its sum: split $a_n$ into positive and negative parts $a_n^{\pm} = \max(\pm a_n, 0)$, both summable, and apply the proposition to each. By contrast, a conditionally convergent series can be rearranged to have any sum (Riemann).

**Theorem 2.17 (2.4.2, alternating series test)**

If $x_1 \geq x_2 \geq \cdots \geq 0$ and $x_n \to 0$, then $\sum (-1)^{n+1} x_n$ converges, and its sum $s$ satisfies $|s - s_n| \leq x_{n+1}$.

**Proof.** $s_{2m} = (x_1 - x_2) + \cdots + (x_{2m-1} - x_{2m})$ increases, $s_{2m+1} = x_1 - (x_2 - x_3) - \ldots$ decreases, and $s_{2m} \leq s_{2m+1}$. Both converge (Theorem 1.15), and $s_{2m+1} - s_{2m} = x_{2m+1} \to 0$, so they have the same limit $s$, which lies between consecutive partial sums. ■

**Theorem 2.18 (2.4.3, comparison test)**

If $0 \leq a_n \leq b_n$ and $\sum b_n$ converges, then $\sum a_n$ converges.

**Proof.** The partial sums of $\sum a_n$ are bounded by $\sum b_n$; use Corollary 1.16. ■

**Proposition 2.19 (2.4.4, root test)**

Let $\alpha = \limsup |a_n|^{1/n}$. If $\alpha < 1$, $\sum a_n$ converges absolutely; if $\alpha > 1$, $\sum a_n$ diverges.

**Proof.** If $\alpha < \beta < 1$, then $|a_n| < \beta^n$ for large $n$ (Proposition 1.17), and $\sum \beta^n < \infty$. If $\alpha > 1$, $|a_n| > 1$ for infinitely many $n$, so $a_n \not\to 0$. ■

**Proposition 2.20 (2.4.5, ratio test)**

Let $a_n \neq 0$. If $\limsup |a_{n+1}/a_n| < 1$, then $\sum a_n$ converges absolutely; if $|a_{n+1}/a_n| \geq 1$ for all large $n$, then $\sum a_n$ diverges.

**Proof.** If $|a_{n+1}/a_n| \leq \beta < 1$ for $n \geq N$, then $|a_n| \leq |a_N|\beta^{n-N}$. In the second case $|a_n|$ is eventually nondecreasing and positive, so $a_n \not\to 0$. ■

**Proposition 2.21 (2.4.6)**

If $c_n > 0$, then $\liminf \frac{c_{n+1}}{c_n} \leq \liminf c_n^{1/n} \leq \limsup c_n^{1/n} \leq \limsup \frac{c_{n+1}}{c_n}$.

**Proof.** Let $\beta = \limsup c_{n+1}/c_n$, assumed finite, and $\gamma > \beta$. For $n \geq N$, $c_{n+1} < \gamma c_n$, so $c_n < c_N \gamma^{-N} \gamma^n$ and $c_n^{1/n} < (c_N \gamma^{-N})^{1/n} \gamma \to \gamma$. Hence $\limsup c_n^{1/n} \leq \gamma$ for all $\gamma > \beta$. The left inequality is analogous. ■

So the root test is stronger than the ratio test: whenever the ratio test decides, so does the root test.

## 2.5 Compact sets

**Motivation.** A closed bounded set in $\mathbb{R}^n$ has the property that every open cover has a finite subcover. This abstract property is what lets us pass from local estimates to global ones, and Heine–Borel shows it is the same as being closed and bounded.

An **open cover** of $K$ is a family of open sets whose union contains $K$; $K$ is **compact** if every open cover has a finite subcover.

**Proposition 2.22 (2.5.1)**

Every compact set is bounded and closed.

**Proof.** The sets $N(0,k)$ cover $K$; finitely many do, so $K \subset N(0,m)$. If $x \notin K$, the open sets $U_k = \{y : \|y - x\| > 1/k\}$ cover $K$; a finite subcover gives $K \subset U_m$, so $N(x, 1/m)$ misses $K$. Hence $\mathbb{R}^n - K$ is open. ■

**Theorem 2.23 (2.5.2)**

Every box $B = I_1 \times \cdots \times I_n$ is compact.

**Proof.** Suppose an open cover $\{U_i\}$ of $B$ has no finite subcover. Bisecting all sides, one of the $2^n$ sub-boxes, $B_2$, is not covered by finitely many $U_i$; continuing gives nested boxes $B_m$ of diameter $\operatorname{diam} B_1/2^{m-1}$ none finitely covered. Let $x \in \bigcap B_m$ (Lemma 2.7) and $x \in U_i$, $N(x,\epsilon) \subset U_i$. For large $m$, $B_m \subset N(x,\epsilon) \subset U_i$, a contradiction. ■

**Proposition 2.24 (2.5.3)**

A closed subset $F$ of a compact set $K$ is compact.

**Proof.** If $\{U_i\}$ covers $F$, then $\{U_i\} \cup \{\mathbb{R}^n - F\}$ covers $K$; a finite subcover, with $\mathbb{R}^n - F$ removed, covers $F$. ■

**Theorem 2.25 (2.5.4, Heine–Borel)**

For $K \subset \mathbb{R}^n$ the following are equivalent:
(a) $K$ is compact;
(b) $K$ is bounded and closed;
(c) every infinite subset of $K$ has a limit point in $K$.

**Proof.** (a)⇒(b): Proposition 2.22. (b)⇒(a): $K$ lies in a box, which is compact (Theorem 2.23); apply Proposition 2.24. (b)⇒(c): Bolzano–Weierstrass gives a limit point, which is in $K$ as $K$ is closed. (c)⇒(b): if $K$ were unbounded, choose $x_k \in K$ with $\|x_k\| > k$; the infinite set $\{x_k\}$ has no limit point at all. If $x \in K'$, Proposition 2.9 gives distinct $x_k \in K$ with $x_k \to x$; the set $\{x_k\}$ has $x$ as its only limit point, so $x \in K$ by (c), and $K$ is closed. ■

So $K$ is compact iff every sequence in $K$ has a subsequence converging in $K$.

**Theorem 2.26 (2.5.5, finite intersection property)**

If $\{K_i : i \in I\}$ are compact sets such that $\bigcap_{i\in J} K_i \neq \emptyset$ for every finite $J \subset I$, then $\bigcap_{i\in I} K_i \neq \emptyset$.

**Proof.** Fix $i_0$. If the intersection were empty, the open sets $\mathbb{R}^n - K_i$ would cover $K_{i_0}$, and finitely many $\mathbb{R}^n - K_{i_1}, \dots, \mathbb{R}^n - K_{i_m}$ would cover it; then $K_{i_0} \cap K_{i_1} \cap \cdots \cap K_{i_m} = \emptyset$, a contradiction. ■

In particular a decreasing sequence of nonempty compact sets has nonempty intersection.

**Corollary 2.27 (2.5.6)**

A nonempty countable closed set $S \subset \mathbb{R}^n$ has an isolated point.

**Proof.** Let $S = \{s_1, s_2, \dots\}$ and suppose no point is isolated. Choose a closed ball $K_1 = \overline{N(y_1, r_1)}$ with $y_1 \in S$; inductively, since $y_k$ is not isolated, $N(y_k, r_k)$ contains infinitely many points of $S$, so we can choose $y_{k+1} \in S \cap N(y_k, r_k)$, $y_{k+1} \neq s_k$, and a closed ball $K_{k+1}$ about $y_{k+1}$ inside $N(y_k, r_k)$ not containing $s_k$. The sets $K_k \cap S$ are nonempty, compact and decreasing, so they have a common point (Theorem 2.26); it lies in $S$ but differs from every $s_k$, a contradiction. ■

Thus a nonempty closed set without isolated points (a **perfect set**), such as the Cantor set, is uncountable.

## 2.6 Connected sets

**Motivation.** Intervals are the sets that cannot be split into two separated pieces. This property is what lies behind the intermediate value theorem.

$S \subset \mathbb{R}^n$ is **disconnected** if there are open $U, V$ with

$$U \cap S \neq \emptyset, \quad V \cap S \neq \emptyset, \quad U \cap V \cap S = \emptyset, \quad S \subset U \cup V,$$

and **connected** otherwise. A set $S \subset \mathbb{R}$ is an **interval** if $x, y \in S$ and $x < z < y$ imply $z \in S$.

**Proposition 2.28 (2.6.1)**

A subset $S$ of $\mathbb{R}$ is connected iff it is an interval.

**Proof.** If $S$ is not an interval, take $x < z < y$ with $x, y \in S$, $z \notin S$; then $U = (-\infty, z)$, $V = (z, \infty)$ disconnect $S$. Conversely let $S$ be an interval and $U, V$ as above, with $x \in U \cap S$, $y \in V \cap S$, say $x < y$; then $[x, y] \subset S$. Let $c = \sup([x, y] \cap U)$. If $c \in U$, then $c < y$ (as $y \notin U$) and $U$ contains $[c, c+\delta)$, contradicting the definition of $c$; if $c \in V$, then $c > x$ and $(c-\delta, c] \subset V$, so $(c-\delta, c]$ misses $U$ and $c$ is not the supremum. But $c \in [x, y] \subset S \subset U \cup V$, a contradiction. ■

**Proposition 2.29 (2.6.2)**

If $\{C_i\}$ are connected and $\bigcap C_i \neq \emptyset$, then $C = \bigcup C_i$ is connected.

**Proof.** Suppose $U, V$ disconnect $C$, and let $p \in \bigcap C_i$, say $p \in U$. Each $C_i$ is connected and meets $U$ (at $p$), so $C_i \cap V = \emptyset$ (otherwise $U, V$ would disconnect $C_i$). Then $C \cap V = \emptyset$, a contradiction. ■

Convex sets, in particular balls and boxes, are connected: they are unions of segments through a common point, each segment being connected as a continuous image of an interval (Theorem 3.13).